

Midterm 1 instructor guide

Slide route: Syllabus date: October 26, 2026. Proposed coverage: Lectures 1–4, sessions 1–5.

Session 6 | Midterm 1

Purpose: Assessment session, with no new examinable material. Date from the syllabus: October 26. Position in the 16-session sequence is provisional until the meeting calendar is checked.

Before the day Confirm room, duration, permitted materials, accessibility arrangements, and the final paper through the course's usual procedures. This guide is a proposed assessment blueprint, not an approved exam or a change to policy.

Suggested assessment balance

Area	Share	What it should reveal
Probability	20%	Joint versus conditional probabilities; iterated expectations.
Sampling	20%	Mean, variance, LLN versus CLT.
Inference	25%	Null, statistic, decision, interpretation.
Regression	35%	OLS arithmetic, exogeneity, robust SEs.

Step-by-step exam-day script

Step 1 | Arrival and instructions

Guide Welcome students, check materials and accessibility arrangements, and state the time clearly. Do not add content beyond the approved instructions.

Step 2 | Probability question

Guide Use a joint table or conditional-probability question. Require students to identify the reference population before calculating.

Step 3 | Moments question

Guide Ask for a mean and variance, preferably in two equivalent forms. Award the calculation only when units and probabilities are clear.

Step 4 | Sampling question

Guide Test the distinction between LLN and CLT. A good answer says concentration is not the same as normality.

Step 5 | Inference question

Guide Require a null, statistic, reference distribution, decision, and interpretation. Penalize a p-value described as the probability that the null is true.

Step 6 | Regression arithmetic

Guide Use a short OLS line or residual calculation. Check that students distinguish population errors from sample residuals.

Step 7 | Causal interpretation

Guide Include an assumption question about conditional mean zero or random assignment. Arithmetic alone should not earn full credit.

Step 8 | Standard errors

Guide Ask whether robust SEs alter coefficients or repair endogeneity. The answer is no to both.

Step 9 | Review and submission

Guide Give a five-minute warning, then collect all pages and record timing or wording issues for later review.

100-minute slot assumption Allow 10 minutes for settling and instructions, 80 minutes of assessment, and 10 minutes for collection. Replace this allocation if the official exam duration differs.

Instructions to emphasize Define notation, show calculations, state the null, identify the reference distribution, and distinguish a statistical conclusion from a causal claim. Clarify permitted resources from the actual exam instructions.

Marking principle Separate arithmetic from interpretation. Give appropriate partial credit for a correct method after a carried-forward numerical error. Use the same standard across scripts.

Pre-exam oral check | With answers

A continuous variable has zero probability at a point: intervals can still have positive probability. A mean's SE falls at rate $1/\sqrt{n}$ under i.i.d. finite-variance sampling. OLS residual orthogonality does not establish exogeneity. Robust SEs do not repair an endogenous regressor.

After collection Record unclear wording or timing problems without changing grading ad hoc. Prepare a short diagnostic for the next session; avoid announcing a universal weakness before inspecting the scripts.

Next session: Multiple regression and omitted-variable bias. Begin with a five-minute distinction between fit, uncertainty, and causal interpretation.